CSC4197

RESEARCH METHODOLOGY

WRITING TOOLS — LATEX, OVERLEAF & MS WORD



2 WHY TOOLING MATTERS

- A research paper is a **structured document**: numbered sections, cross-referenced figures/tables/equations, auto-generated bibliography
- Manual formatting = wasted hours + broken references at 2 a.m. before the deadline
- Two professional toolchains:
  - **LaTeX** (the CS standard)
  - **MS Word** (used correctly — with styles & fields, not manual formatting)

3 WHAT IS LATEX?

- A **markup-based typesetting system**: you write plain text with commands; the engine produces a polished PDF
- Separation of **content** from **presentation** — the venue's class file handles all formatting
- Strengths: math, cross-references, bibliographies, version control, two-column layouts, stability for 100+ page theses
- **Overleaf** = browser-based LaTeX with real-time collaboration (Google-Docs-style) — no installation needed
- **Prism** = May be an alternative for overleaf, more features in free version than overleaf

4 LATEX DOCUMENT SKELETON

- `\documentclass[conference]{IEEEtran}`
- `\usepackage{graphicx, amsmath, booktabs}`
- `\begin{document}`
- `\title{Your Paper Title}`
- `\author{\IEEEauthorblockN{Student Name}`
- `\IEEEauthorblockA{Dept. of CS, AIUB, Dhaka, Bangladesh}}`
- `\maketitle`
- `\begin{abstract}...\end{abstract}`
- `\section{Introduction}\label{sec:intro}`
- `As shown in Section~\ref{sec:method}...`
- `\section{Methodology}\label{sec:method}`
- `\bibliographystyle{IEEEtran}`
- `\bibliography{refs}`
- `\end{document}`
- One template, infinite papers.

5 LATEX ESSENTIALS YOU'LL USE WEEKLY

- **Math:** `$O(n \log n)$` → O(n log n); display equations with `\begin{equation}`
- **Figures:**
- `\begin{figure}[t]`
- `  \includegraphics[width=\columnwidth]{results.pdf}`
- `  \caption{F1 vs. training size.}\label{fig:f1}`
- `\end{figure}`
- **Tables:** use `booktabs` (`\toprule \midrule \bottomrule`) — never vertical lines
- **Cross-refs:** `\label{}` + `\ref{}` — numbers update automatically
- **Algorithms:** `algorithm2e` / `algorithmicx` packages
- **Citations:** `\cite{rahman2024}` (full workflow next week)

6 MS WORD, THE PROFESSIONAL WAY

- Word is fine **if** you use its structural features:
- **Styles** (Heading 1/2/3, Caption) — never manual bold+font-size
- **Multilevel numbering** bound to heading styles
- **Insert** → **Caption** for figures/tables + **Cross-reference** fields
- **References tab** — citation manager integration (Zotero plugin, Week 9)
- **Automatic ToC** from styles (essential for thesis)
- **Track Changes + Comments** — how supervisors will review your thesis drafts
- Equation editor (`Alt + =`), Navigation pane for structure

7 LATEX VS. WORD: CHOOSING

Criterion	LaTeX	Word
Math-heavy content	✅ superior	workable
Venue templates (IEEE/ACM/Springer)	✅ native	sometimes provided
Long thesis stability	✅	risky (corruption, ref breakage)
Track-changes review with non-tech supervisor	harder	✅
Learning curve	steeper	shallow
Collaboration	Overleaf ✅	OneDrive/Docs ✅

Rule 1: the venue decides. If IEEE conference → LaTeX template. University thesis office wants .docx → Word (with styles).

8 COLLABORATIVE & VERSIONED WRITING

- **Overleaf sharing:** edit links, comments, track changes (paid), version history
- **Git for papers:** `paper.tex` in a repo — meaningful commits ("add ablation table"), branches for major rewrites; Overleaf has GitHub sync
- Golden rules:
  - One sentence per line in source → clean diffs
  - Figures generated by **scripts** (matplotlib → PDF) committed alongside
  - Never email `final_v7_REAL_final.docx` — that workflow ends today

9 CASE STUDY: THE CAMERA-READY CRISIS

- **True-to-life scenario:** A team writes their accepted paper in Word with manual numbering. Camera-ready requires the IEEE LaTeX template, a new reference style, and a 6-page hard limit — due in 72 hours.
- Manual references must all be rebuilt → 6 hours lost
- Figures were low-res PNG screenshots → fail the PDF compliance check
- Manual "Figure 5" mentions now point to the wrong figures
- **Lesson:** start in the venue's template from day one; scriptable figures; automated references. Discussion: which of these mistakes is most expensive?

10 HANDS-ON LAB PART A (25 MIN): OVERLEAF

- Follow along — by the end you will have:
- Created a project from the **IEEE conference template**
- Set your project's real title/authors/abstract (from Week 7 workshop!)
- Added Section headings matching IMRaD
- Inserted one equation, one `booktabs` table, one figure with caption + `\ref`
- Compiled to PDF and shared the project link with teammates + instructor
- Stuck? The error log line number is your friend — read the **first** error only.

11 HANDS-ON LAB PART B (15 MIN): WORD DONE RIGHT

- Using the provided thesis-chapter starter file:
- Apply Heading 1/2 styles; watch the Navigation pane organize itself
- Insert a captioned figure + a cross-reference to it
- Generate an automatic Table of Contents
- Turn on Track Changes; make an edit; add a comment; accept/reject

12 WRAP-UP & NEXT WEEK

- **Key takeaways**
- Venue template decides your tool; learn both toolchains
- Structure (styles/commands) over manual formatting — always
- Your project paper now lives in Overleaf — all future writing happens there
- **Next week:** Reference management — Zotero, BibTeX, citation styles; bring your Assignment 1 paper PDFs
- **Install before class:** Zotero + browser connector (free)